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Introducing PTX12000 Chassis

An HPE Juniper Networking Techpost

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1 Introducing PTX12000 Chassis

Nicolas Fevrier - 02/25/2026

- Introducing the PTX12000 Series (this blog)
- Rethinking Fabric Redundancy
- Deepdive in the Fabric Topology
- Fabric Self Healing and Recovery
- Navigating the Front Panel Display
- PTX12k Power Saving and Metrics (coming soon)

1.1 Introduction

It's not every day you unveil a brand-new series product line, so it's with a lot of excitement we launched the PTX12000 modular chassis in February 2026.

The PTX12000 chassis sets new standards in terms of port density (800GbE today, 1.6TE will come next), cooling capacity and power efficiency. The new chassis completes the PTX10000 family by adding a different form factor and a fully redesigned architecture. Both the latest line cards and fabric cards for PTX10000 and PTX12000 use the same internal packet-forwarding engine: the Express 5 ASICs.

The new system will be released in two form factors:

- An 8-slot chassis that measures 22RU and is already shipping
- A 12-slot version, 32RU, planned for the end of the calendar year 2026 (contact your favorite HPE representative for the latest details).

And day 1, they can be populated with two types of line cards:

- 54 ports 800GbE OSFP
- 54 ports 800GbE QSFP-DD

This article will provide an extensive description of the hardware (line cards, fabric cards, routing boards, fan trays and power modules), but we will cover also the interfaces and the port capabilities.

Many other publications will follow and will be linked from this page when they appear online.

As a prerequisite, for more details on the Packet Forwarding Engine and the Express5 variants, we invite you to read the following blog posts first:

- Introduction to Express 5 TechPost blog: https://techpost.hpe.com/articles/express-5-overview/article
- https://techpost.hpe.com/articles/introducing-express5-in-ptx10k-chassis/article (section "Main Forwarding Components")

A bit of vocabulary. In this document, we will talk about:

- "Package": to describe the Express 5 ASIC. Alternatively, we will use "chipset", "NPU" (for Network Processing Unit) or even "Forwarding Engine".
- "PFE-instance" to describe the BX chiplet, made of two datapath (DPs)
- "PFE" to describe a 7.2Tbps datapath in the PFE-instance.

1.2 Chassis Description

In this section, we'll illustrate the description with details on the 8-slot chassis, but all these concepts apply to the 12-slot version too, unless otherwise specified.

1.3 Front View

Figure 02 above provides a front view of the chassis in diverse situations:

- Left: with the EMI protective door installed
- Center: with line cards and routing engine inserted
- Right: without any FRU (Field Replaceable Unit) installed, except the fan trays and one fabric card.

Let's describe the chassis starting from the top.

In front of the power shelf (we will cover it later), you'll find a "Front Panel Display". It's a touch screen an on-site operator can use to visualize the various alarms and verify the chassis parts and interfaces states.

Below the display, two slots are available for the Routing and Control Boards (RCB), also called "Routing Engines" (RE). The system can be operated with one or two REs present. They handle the control and management planes functions, but they also control and monitor the entire chassis internals. If you operate the system with two REs, they become fully redundant (active/standby).

Below the REs, we have 8 or 12 horizontal slots for 2.75 inches line cards. At the date of this article creation, two flavors of line cards are available, both offering 54 ports 800GbE with different optics form-factors. It represents 43.2Tbps of port connectivity but also 43.2Tbps of forwarding capability (the cards are not oversubscribed). More line cards will be introduced in the future, and the system is designed to support 1.6T optics.

The following chart describes the maximum port density per interface type/speed at launch day:

Pluggable Optics Modules	Maximum Port Densityper Slot	Maximum Port Densityper PTX12008	Maximum Port Densityper PTX12012
800Gbps	54	432	648
400Gbps	108 (w/ 2x400G)	864	1,296
100Gbps	432 (w/ 8x100G)	3,456	5,184
40Gbps	54 (QSFP only)	432	648
25Gbps	216 (w/ 4x25G, QSFP only)	1,728	2,592
10Gbps	216 (w/ 4x10G, QSFP only)	1,728	2,592

From a power and cooling perspective, it's possible to operate high-power optics ZR/ZRP on all ports with no restriction: 30W per QSFP port and 33W per OSFP port.

Note that you can only insert “PTX12K-LC” line cards in these slots (PTX12K-LC54OSFP/PTX12K-LC54QDD today, more in the future), but the PTX10000 cards, even those based on Express5, are not compatible. It’s a totally different pitch, and different number of connectors to the fabric.

Below the line cards slots, at the bottom of the chassis, you’ll find the ESD ground point:

1.4 Back View

At the top, you’ll find a power shelf. For the PTX12008, this shelf can host up to 20 (4x5) Power Supply Units (PSU). The PSU are also seen as Power Supply Modules (PSM) from the operating system’s perspective. The PTX12012 has a higher power shelf (4x9).

Two PSM options are available at the moment of the publication of this article:

- JNP-PWR-3K-AC PSU: dual-feed 3,000W AC, High Voltage HVAC/HVDC
- JNP-PWR-3K-DC PSU: dual-feed 3,000W DC that can be operated at 60A or 80A

Mixing AC and DC in the same power shelf is not supported.

Below the PTX12008 shelf, you can see three large fan trays, each made of 10 (5x2) large counter-rotating fans. They guarantee the front-to-back cooling of the system and offer 2+1 redundancy.

Behind each fan tray, 3x fabric cards are inserted vertically. We have a total of 9x fabric cards in the chassis (both PTX12008 and PTX12012, but of course, the fabric cards are different for each chassis type). To operate, remove or insert, a fabric card, you need to unseat the facing fan tray. Please refer to the hardware guide for details on the operation.

From the back of the router, the ESD attach point and the earth ground points are located on the right of the fan trays.

1.5 Let’s Verify That on the Console

The following show command output displays the various parts of a PTX12008 with 2x RCBs, 8x Line Cards (QSFP and OSFP), 9x Fabric Cards, 3x Fan Trays, and 20x Power Supply Modules inserted in the shelf.

```
root@ptx12008-re0> show chassis hardware detail
Hardware inventory:
Item Version Part number Serial number Description
Chassis FX032 PTX12008
Midplane 0 REV 01 750-180225 BCFZ3803 Midplane 8
FPM 0 REV 01 750-181246 BCGB4121 Front Panel Display
PSM 0 REV 05 740-143236 1GH1D130494 JNP 3000W HVAC/HVDC Power Supply
PSM 1 REV 05 740-143236 1GH1D130478 JNP 3000W HVAC/HVDC Power Supply
PSM 2 REV 05 740-143236 1GH1D130499 JNP 3000W HVAC/HVDC Power Supply
PSM 3 REV 05 740-143236 1GH1D130486 JNP 3000W HVAC/HVDC Power Supply
PSM 4 REV 05 740-143236 1GH1D130487 JNP 3000W HVAC/HVDC Power Supply
PSM 5 REV 05 740-143236 1GH1D130483 JNP 3000W HVAC/HVDC Power Supply
PSM 6 REV 05 740-143236 1GH1D130463 JNP 3000W HVAC/HVDC Power Supply
PSM 7 REV 05 740-143236 1GH1D130474 JNP 3000W HVAC/HVDC Power Supply
PSM 8 REV 05 740-143236 1GH1D130469 JNP 3000W HVAC/HVDC Power Supply
PSM 9 REV 05 740-143236 1GH1D130485 JNP 3000W HVAC/HVDC Power Supply
PSM 10 REV 06 740-143236 1GH1E430638 JNP 3000W HVAC/HVDC Power Supply
PSM 11 REV 06 740-143236 1F36E500027 JNP 3000W HVAC/HVDC Power Supply
PSM 12 REV 06 740-143236 1F36E500126 JNP 3000W HVAC/HVDC Power Supply
PSM 13 REV 06 740-143236 1F36E500007 JNP 3000W HVAC/HVDC Power Supply
PSM 14 REV 06 740-143236 1F36E500012 JNP 3000W HVAC/HVDC Power Supply
PSM 15 REV 05 740-143236 1GH1D130429 JNP 3000W HVAC/HVDC Power Supply
PSM 16 REV 05 740-143236 1GH1D130489 JNP 3000W HVAC/HVDC Power Supply
PSM 17 REV 05 740-143236 1GH1D130491 JNP 3000W HVAC/HVDC Power Supply
PSM 18 REV 05 740-143236 1GH1D130476 JNP 3000W HVAC/HVDC Power Supply
```

```
PSM 19 REV 05 740-143236 1GH1D130498 JNP 3000W HVAC/HVDC Power Supply
Routing Engine 0 BUILTIN BUILTIN PTX12K-RE
  nvme0 400088 MB VTPM28ADCI400-BJ230005 63871-0006 NVMe Solid State Disk
  nvme1 400088 MB VTPM28ADCI400-BJ230005 64442-0013 NVMe Solid State Disk
Routing Engine 1 BUILTIN BUILTIN PTX12K-RE
  nvme0 400088 MB DSM28-400DP1KWAEFP-B553B Y5342501160070119 NVMe Solid State Disk
  nvme1 400088 MB DSM28-400DP1KWAEFP-B553B Y5342501160070107 NVMe Solid State Disk
CB 0 REV 06 750-173945 BCFS1429 Control Board
CB 1 REV 12 750-173945 BCGR4043 Control Board
FPC 0 REV 29 750-173325 BCGJ3200 PTX12K-LC54QDD
  CPU REV 05 750-173947 BCGF9527 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54QDD800-LC-PIC
    Xcvr 0 REV 01 740-150874 1W1CSMAA07004 QSFP-DD800-800G-AOC-10M
    ...
    Xcvr 53 REV 01 740-150874 1W1CSMAA0700D QSFP-DD800-800G-AOC-10M
  MEZZ 0 REV 09 750-175158 BCGF9428 PTX12K-LC54QDD MEZZ A Board
  MEZZ 1 REV 08 750-175159 BCGG0265 PTX12K-LC54QDD MEZZ B Board
  MEZZ 2 REV 06 750-175033 BCGF5395 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 07 750-175031 BCGF4361 PTX12K-LC54 Power Board Left
FPC 1 REV 17 750-173322 BCFZ3747 PTX12K-LC54OSFP
  CPU REV 04 750-173947 BCFZ6313 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
    Xcvr 0 REV 01 740-174938 1G1THKA95200Z OSFP-8x100G-FR1
    Xcvr 13 REV 01 740-174938 1G1THKA95100L OSFP-8x100G-FR1
    Xcvr 46 REV 01 740-174935 1W1CVXAA0400M OSFP-2x400G-FR4-P
  MEZZ 0 REV 04 750-175154 BCFZ6245 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 04 750-175157 BCFZ7496 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 04 750-175033 BCFX9225 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 05 750-175031 BCFZ2121 PTX12K-LC54 Power Board Left
FPC 2 REV 29 750-173325 BCGJ3197 PTX12K-LC54QDD
  CPU REV 05 750-173947 BCGF9519 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54QDD800-LC-PIC
    Xcvr 0 REV 01 740-150875 1W1CSNAA08008 QSFP-DD800-800G-AOC-15M
    ...
    Xcvr 53 REV 01 740-150876 1W1CSOAA04003 QSFP-DD800-800G-AOC-20M
  MEZZ 0 REV 09 750-175158 BCGF9424 PTX12K-LC54QDD MEZZ A Board
  MEZZ 1 REV 08 750-175159 BCGG0257 PTX12K-LC54QDD MEZZ B Board
  MEZZ 2 REV 06 750-175033 BCGF5402 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 07 750-175031 BCGF4365 PTX12K-LC54 Power Board Left
FPC 3 REV 29 750-173322 BCGJ3189 PTX12K-LC54OSFP
  CPU REV 05 750-173947 BCGF9523 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
  MEZZ 0 REV 09 750-175154 BCGF9247 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 08 750-175157 BCGF9432 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 06 750-175033 BCGF5418 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 07 750-175031 BCGF4376 PTX12K-LC54 Power Board Left
FPC 4 REV 17 750-173322 BCFY2429 PTX12K-LC54OSFP
  CPU REV 04 750-173947 BCFZ6306 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
    Xcvr 22 REV 01 740-174938 1G1THKA95200B OSFP-8x100G-FR1
    Xcvr 37 REV 01 740-174933 1A1MVWA9400EZ OSFP-800G-DR8-P
    Xcvr 43 REV 01 740-174933 1A1MVWA94008L OSFP-800G-DR8-P
  MEZZ 0 REV 04 750-175154 BCFZ6255 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 04 750-175157 BCFZ7448 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 04 750-175033 BCFX9240 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 05 750-175031 BCFZ2111 PTX12K-LC54 Power Board Left
FPC 5 REV 46 750-173322 BCGT8620 PTX12K-LC54OSFP
  CPU REV 08 750-173947 BCGT9286 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
  MEZZ 0 REV 12 750-175154 BCGV2215 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 10 750-175157 BCGT9230 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 08 750-175033 BCGV2153 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 09 750-175031 BCGV3115 PTX12K-LC54 Power Board Left
FPC 6 REV 17 750-173322 BCFY2414 PTX12K-LC54OSFP
  CPU REV 04 750-173947 BCFZ6346 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
    Xcvr 5 REV 01 740-174933 1A1MVWAA160CS OSFP-800G-DR8-P
...
    Xcvr 53 XXXX NON-JNPR 31360 OSFP-PASSIVE-E-LPBK
  MEZZ 0 REV 04 750-175154 BCFZ6263 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 04 750-175157 BCFZ7453 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 04 750-175033 BCFX9243 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 05 750-175031 BCFZ1900 PTX12K-LC54 Power Board Left
FPC 7 REV 17 750-173322 BCFY2416 PTX12K-LC54OSFP
  CPU REV 04 750-173947 BCFZ6318 PTX12K-LC54 PMB Board
  PIC 0 BUILTIN BUILTIN PTX12K-54OSFP800-LC-PIC
  MEZZ 0 REV 04 750-175154 BCFZ6281 PTX12K-LC54OSFP MEZZ A Board
  MEZZ 1 REV 04 750-175157 BCFZ7499 PTX12K-LC54OSFP MEZZ B Board
  MEZZ 2 REV 04 750-175033 BCFX9221 PTX12K-LC54 Power Board Right
  MEZZ 3 REV 05 750-175031 BCFZ2102 PTX12K-LC54 Power Board Left
SIB 0 REV 20 750-173308 BCGR7353 SIB-PTX12008-SF
```

```
  MEZZ 0 REV 04 750-176911 BCGS0396 SIB-PTX12008-SF Power Board
SIB 1 REV 20 750-173308 BCGR7344 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGS0406 SIB-PTX12008-SF Power Board
SIB 2 REV 20 750-173308 BCGR7339 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGS0379 SIB-PTX12008-SF Power Board
SIB 3 REV 22 750-173308 BCGV5056 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGV2204 SIB-PTX12008-SF Power Board
SIB 4 REV 20 750-173308 BCGR7335 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGS0407 SIB-PTX12008-SF Power Board
SIB 5 REV 22 750-173308 BCGV5055 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGV2209 SIB-PTX12008-SF Power Board
SIB 6 REV 22 750-173308 BCGV5061 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGV2196 SIB-PTX12008-SF Power Board
SIB 7 REV 22 750-173308 BCGV5059 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGV2164 SIB-PTX12008-SF Power Board
SIB 8 REV 22 750-173308 BCGV5058 SIB-PTX12008-SF
  MEZZ 0 REV 04 750-176911 BCGT2281 SIB-PTX12008-SF Power Board
Fan Tray 0 REV 01 750-175846 BCFW3852 PTX12008 Fan tray
Fan Tray 1 REV 01 750-175846 BCFW3820 PTX12008 Fan tray
Fan Tray 2 REV 01 750-175846 BCFW3816 PTX12008 Fan tray
POWERSHELF 0 BUILTIN BUILTIN 5 Row PSM POWERSHELF

{master}
root@ptx12008-re0>
```

1.6 Dimension/Weight/Installation

With all Field Replacement Units (FRU) removed, an empty chassis weights 325 lb (147.4 kg). The chassis depth will vary depending on the cosmetics installed, cable management and the protective front door, as displayed in Figure 06 below.

Always refer to the hardware guide prior to the installation of the chassis in your facilities. The “Fast Track to Rack Installation” section will provide all the necessary details.

1.7 Chassis Architecture

The PTX12000 chassis uses an “orthogonal-direct” (OD) architecture. In this layout, line cards slide in horizontally at the front of the chassis and connect straight to the fabric cards that sit vertically in its center. Because of this arrangement, no intermediate back-plane or mid-plane is needed. Each Packet Forwarding engine (PFE) links to every fabric engine on all fabric cards, creating a full-mesh interconnection.

Many hardware innovations have been brought during the building of this new chassis, starting from the power distribution. We will cover them in further details in follow up publications.

1.8 Line Cards High-Level Description

The section “Line Card Architecture” will go in deeper internal details, block diagrams and port capabilities, but first, let’s have a quick overview.

The line cards are inserted horizontally on the front side of the chassis and are connected to all the fabric cards (SIBs) present in the center. Nine SIBs are required to reach the full forwarding capacity. The combo LC/SIB forms an internal full mesh, or Clos topology, where every ingress port is one hop away from each egress port.

In Feb 2026, when we launch the PTX12000 chassis, two interface cards are available:

- PTX12K-LC54QDD

- PTX12K-LC54OSFP

If you have been in this industry for long enough, the first thing you'll notice is the height (or pitch) of these new cards. In a PTX10000 chassis, for example, the cards are slightly less than 5cm high, and the usual range you'll find in the market spans from 4.5 to 5.4cm. The two LC54 cards are **7cm** high or **2.75 inches**. This unique form-factor allows higher port density (54 interfaces on 3 rows) and improved cooling capacity.

The cards can be positioned in any slot without any restriction, mixed and matched as you wish. Powered by three Express 5 BXF Packet Forward Engine for 43.2Tbps of forwarding capacity, they offer the highest port density per card in the industry: 54x 800Gbps ports (and yes, we mean real 800GbE here, but also 2x 400GbE, 8x 100GbE and many other channelization/breakout options).

They are hot removable and insertable, without any specific impact on the chassis. The use of a specific CLI "request chassis fpc slot X offline/online" is required for a smooth operation.

The benefits of such a unique hardware architecture and port density will be elaborated in dedicated blog posts. Without spoiling them nor going into too many details, I want to emphasize some key aspects:

- It's a front-to-back cooling design, the cool air flow is received from the optics and goes towards the back of the chassis but also through the power shelf
- All ports, QSFP or OSFP, can host high power 800G optics (ZR, ZR+). Cooling and power budget is designed to support all ports, on all slots, with 30W and 33W optics.
- All ports can be encrypted at L2, line rate, with MACsec technology

The hardware guide will describe the cosmetics installation, cable management, etc.

Note that in normal operational conditions, the card is not equipped with ejector tool handles, they are only required for OIR (insertion/ejection). Here again, we invite you to carefully read the hardware guide for the exact procedure.

Considering the weight of the line cards (21kg of OSFP and 21.3kg and QSFP), we recommend two operators onsite or the use of specific lifting equipment to manipulate it.

1.9 Fabric Cards

The PTX12000 chassis can be populated with a maximum of 9 fabric cards named "PTX12008-SF" and "PTX12012-SF".

This number of fabric cards is required to offer the maximum possible bandwidth per PFE/Datapath, Chipset/-Package and line card. Running the system with less than nine SIBs is fully supported and some commercial bundles rely on this capability (described later in " Running the chassis with less than 9x SIB" section). It's particularly useful if you don't need 43.2Tbps per slot.

The SIBs are inserted from the back of the chassis and occupy a central part in the design. They have 8 of 12 connectors to link to the line cards. Each fabric card is behind a fan tray, and it will be necessary to eject the fans first to insert/remove/replace a SIB.

- SIB 0, 1 and 2 are behind Fan Tray 0
- SIB 3, 4 and 5 are behind Fan Tray 1
- SIB 6, 7 and 8 are behind Fan Tray 2

In these PTX12008-SF, you find 8x connectors to the cards, 6x retimers and 3x Express 5 BF ASICs (check figure 10 below). Each BF chipset can be seen as 144x144 crossbar with 106Gbps links and is designed to switch cells between the BXF PFEs.

```
root@ptx12008-re0> show chassis sibs
Slot State Fabric links Errors
 0 Online Active None
 1 Online Active None
 2 Online Active None
 3 Online Active None
 4 Online Active None
 5 Online Active None
 6 Online Active None
 7 Online Active None
 8 Online Active None

{master}
root@ptx12008-re0> show chassis sibs detail
Slot 0 information:
  State Online
  Uptime 2 days, 1 hour, 19 minutes, 32 seconds
  Fabric links Active
  Errors None
Slot 1 information:
  State Online
  Uptime 2 days, 1 hour, 17 minutes, 8 seconds
  Fabric links Active
  Errors None
Slot 2 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 51 seconds
  Fabric links Active
  Errors None
Slot 3 information:
  State Online
  Uptime 2 days, 1 hour, 17 minutes
  Fabric links Active
  Errors None
Slot 4 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 50 seconds
  Fabric links Active
  Errors None
Slot 5 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 56 seconds
  Fabric links Active
  Errors None
Slot 6 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 54 seconds
  Fabric links Active
  Errors None
Slot 7 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 57 seconds
  Fabric links Active
  Errors None
Slot 8 information:
  State Online
  Uptime 2 days, 1 hour, 16 minutes, 52 seconds
  Fabric links Active
  Errors None

{master}
root@ptx12008-re0>
```

Details on connectivity to the line cards will be covered in a dedicated article: link will come soon :)

1.10 Running the chassis with less than 9x SIB

Operating the system with nine or fewer fabric cards will have a linear impact on the forwarding capability of each PFE/DataPath.

It's essential to understand this aspect: if you are not consuming the entire bandwidth of a DataPath, the remaining forwarding capability is not dynamically re-allocated to a busier PFE. The number of available fabric cards equally impacts all PFEs. This comprehension is key when positioning the interfaces.

SIB	SIB BW per PFE/DP (in Tbps)	BW per slot (in Tbps)
9	7.2	43.2
8	6.4	38.4
7	5.6	33.6
6	4.8	28.8
5	4	24
4	3.2	19.2
3	2.4	14.4
2	1.8	9.6
1	0.8	4.8

Rule of the thumb: the 9 SIBs are necessary for line rate service, and each SIB lost will reduce linearly the forwarding capability by 11%.

Two options are available when ordering the system:

- PTX12008-BASE: **3x SIB**, 1x RE, 12x PSM
- PTX12008-PREMIUM: **9x SIB**, 2x RE, 20x PSM

1.11 Routing and Control Board / Routing Engine

The Routing Engine (SKU: PTX12K-RE-BB/ PTX12K-RE-R) hosts the main compute capabilities of the chassis and centralizes most the control plane and management plane functions, on top of the supervision of the system itself.

Note: the slot in the chassis is marked "RCB" for Routing and Control Board and if the cards are name "RE" they are essentially RCBs.

Component	Description
CPU	Intel Icelake-D 10C @ 2.9GHz
DRAM	256GB DDR4 (4x 64GB DIMM)
SSD	2x 400GB NVMe drives
TPM	Yes, 2.0

Component	Description
Timing	HPE Juniper Timing FPGAConnectivity on the front:1PPS and 10MHz input and outputGPS 1pps input / outputG.703 TOD input portBITS (T1/E1) input / output1G SFP for GM port

This powerful CPU is used by various HPE Juniper software components leveraging multi-core capabilities. For example, Routing Process Daemon (RPD) supports multithreading to process routing updates and routing resolution.

The storage subsystem is comprised of 400GB solid-state drives (SSDs). Two are provided for redundancy, plus reliable management of the software upgrades and rollbacks. These drives are not field-replaceable.

The RE has enough storage, DRAM capacity, and CPU power to host 3rd party applications. Some of these applications may include custom Service Assurance Agents or statistics collection agents developed by HPE Juniper partners and customers. Even full Telegraf, InfluxDB, Grafana stack can run on the router itself for data collection and visualization, check out the blog post (https://techpost.hpe.com/articles/telemetry-collector-and-dataviz-on-junos-evo/article).

From a security perspective, the RE is fully compliant with the TPM2.0 standard as published by the Trusted Computing Group (TCG). The TPM's non-volatile storage is used as a persistent, access-controlled area for component registration, location of policy, keys, etc.Trusted extensions are integrated into the Icelake CPU and only it can access the TPM. Hardware and firmware are designed to support FIPS 140-2 Level 2.

The first release supported on the PTX12008 is Junos Evo 26.2R1. Please reach out to your favorite HPE representative for more details. In Service Software Upgrade (ISSU) is not supported/planned.

1.12 Control Board

The Routing Engine also hosts the control board, necessary for the internal operation of the entire chassis. The details are beyond the scope of this article, but we can say these functions guarantee the internal connectivity from the REs to all the components via an internal 10Gbps Ethernet switched network, but also via PCIe, and I2C networks. These networks are necessary to boot the system, transfer information between components but also monitor and control all conditions during the entire life of the router.

1.13 Front Panel Display

The different line cards have no LED and an operator on site can use the Front Panel Display to inspect the different parameters relative to the chassis, line cards and ports. It's just a display and it's not possible to configure anything or interfere with the operation of the router from this screen.

In a couple of click, it's possible to inspect the state of the different FRUs and ports of the system. A dedicated post will be published on the topic of the FPD navigation. **Link will be added here soon.**

1.14 Front-to-Back Cooling

Multiple fans are used to cool down and preserve the integrity of the system.

- In each PSU active in the power shelf
- In three fan trays installed vertically at the back of the router

These three PTX12008-FAN represent a total of 30 large fans operating simultaneously: 5 pairs of counter-rotating fans per tray (as shown in the output of the “show chassis fan” CLI below).

The PTX12012-FAN have a slightly different design, with 7 large single fans per tray.

Junos constantly collects and monitors dozens of temperature sensors scattered all around the chassis and adjust the rotation speed dynamically to maintain good operational environmental conditions.

```
root@ ptx12008-re0> show chassis environment monitored
Class Item Status Measurement
Power PSM 0 Inlet Temp Sensor OK 24 degrees C / 75 degrees F
     PSM 0 Connector1 Temp Sensor OK 29 degrees C / 84 degrees F
     PSM 0 Connector2 Temp Sensor OK 29 degrees C / 84 degrees F
     PSM 0 Outlet Temp Sensor OK 43 degrees C / 109 degrees F

<SNIP>

root@ptx12008-re0> show chassis environment monitored | count
Count: 1329 lines

{master}

root@ptx12008-re0>
```

The system can operate with one faulty fan tray, the two others will be pushed to full speed. Therefore, it’s possible to extract a fan tray in service to manipulate a SIB.

```
root@ptx12008-re0> show chassis fan
     Item Status % RPM Measurement
     Fan Tray 0 Fan 0 OK 56% 5700 RPM
     Fan Tray 0 Fan 1 OK 62% 7200 RPM
     Fan Tray 0 Fan 2 OK 55% 5550 RPM
     Fan Tray 0 Fan 3 OK 62% 7200 RPM
     Fan Tray 0 Fan 4 OK 55% 5550 RPM
     Fan Tray 0 Fan 5 OK 62% 7200 RPM
     Fan Tray 0 Fan 6 OK 55% 5550 RPM
     Fan Tray 0 Fan 7 OK 62% 7200 RPM
     Fan Tray 0 Fan 8 OK 56% 5700 RPM
     Fan Tray 0 Fan 9 OK 63% 7350 RPM
     Fan Tray 1 Fan 0 OK 56% 5700 RPM
     Fan Tray 1 Fan 1 OK 62% 7200 RPM
     Fan Tray 1 Fan 2 OK 55% 5550 RPM
     Fan Tray 1 Fan 3 OK 62% 7200 RPM
     Fan Tray 1 Fan 4 OK 55% 5550 RPM
     Fan Tray 1 Fan 5 OK 62% 7200 RPM
     Fan Tray 1 Fan 6 OK 55% 5550 RPM
     Fan Tray 1 Fan 7 OK 62% 7200 RPM
     Fan Tray 1 Fan 8 OK 56% 5700 RPM
     Fan Tray 1 Fan 9 OK 61% 7050 RPM
     Fan Tray 2 Fan 0 OK 56% 5700 RPM
     Fan Tray 2 Fan 1 OK 62% 7200 RPM
     Fan Tray 2 Fan 2 OK 55% 5550 RPM
     Fan Tray 2 Fan 3 OK 62% 7200 RPM
     Fan Tray 2 Fan 4 OK 56% 5700 RPM
     Fan Tray 2 Fan 5 OK 62% 7200 RPM
     Fan Tray 2 Fan 6 OK 55% 5550 RPM
     Fan Tray 2 Fan 7 OK 61% 7050 RPM
     Fan Tray 2 Fan 8 OK 55% 5550 RPM
     Fan Tray 2 Fan 9 OK 62% 7200 RPM

{master}
root@ptx12008-re0>
```

1.15 Power

On the top of the router sits a power shelf. It's the same shelf whether we use AC or DC modules. On PTX12008, the shelf can host 20 power supply modules (4x5) while the 12-slot version can accommodate up to 36 units (4x9).

In the figure 14 below, we present the PTX12008 shelf with 20x 3kW DC modules on the left and with AC modules on the right, total 60kW.

These modules convert the grid energy into a 52V current that will be distributed among all the internal FRUs. Both provide dual feed connectors.

Note: The JNP-PWR-3K-AC and JNP-PWR-3K-DC modules are Power Supply Units shared with other product lines like the PTX10000. Also, we can NOT mix AC and DC in the same power shelf. It's one or the other.

The follow show commands provide an exhaustive view of the different PSMs, their feeds, but also the consumption of all internal components in real time (a lot of parts in the chassis, it's a long output).

```
root@ptx12008-re0> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 22370
  PSM 0
    State: Online
    INP0 235 541
    INP1 236 561
    Output 52.49 1005.51
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    INP0 236 559
    INP1 236 544
    Output 52.46 1018.18
    Capacity 3000 W (maximum 3000 W)
  PSM 2
    State: Online
    INP0 236 557
    INP1 236 557
    Output 52.47 1028.21
    Capacity 3000 W (maximum 3000 W)
  PSM 3
    State: Online
    INP0 236 564
    INP1 236 585
    Output 52.43 1078.28
    Capacity 3000 W (maximum 3000 W)
  PSM 4
    State: Online
    INP0 234 551
    INP1 234 558
    Output 52.46 1041.02
    Capacity 3000 W (maximum 3000 W)
  PSM 5
    State: Online
    INP0 235 569
    INP1 235 558
    Output 52.46 1060.77
    Capacity 3000 W (maximum 3000 W)
  PSM 6
    State: Online
    INP0 235 558
    INP1 235 579
    Output 52.48 1051.24
    Capacity 3000 W (maximum 3000 W)
  PSM 7
    State: Online
    INP0 234 538
    INP1 234 558
    Output 52.46 1008.23
    Capacity 3000 W (maximum 3000 W)
  PSM 8
    State: Online
```

```
    INP0 235 561
    INP1 235 551
    Output 52.48 1038.12
    Capacity 3000 W (maximum 3000 W)
  PSM 9
    State: Online
    INP0 235 541
    INP1 235 557
    Output 52.45 1013.07
    Capacity 3000 W (maximum 3000 W)
  PSM 10
    State: Online
    INP0 235 586
    INP1 236 568
    Output 52.47 1085.48
    Capacity 3000 W (maximum 3000 W)
  PSM 11
    State: Online
    INP0 236 560
    INP1 235 558
    Output 52.51 1033.9
    Capacity 3000 W (maximum 3000 W)
  PSM 12
    State: Online
    INP0 235 557
    INP1 234 552
    Output 52.5 1048.51
    Capacity 3000 W (maximum 3000 W)
  PSM 13
    State: Online
    INP0 235 557
    INP1 235 558
    Output 52.46 1036.25
    Capacity 3000 W (maximum 3000 W)
  PSM 14
    State: Online
    INP0 234 551
    INP1 236 555
    Output 52.45 1045.78
    Capacity 3000 W (maximum 3000 W)
  PSM 15
    State: Online
    INP0 234 554
    INP1 234 563
    Output 52.46 1018.14
    Capacity 3000 W (maximum 3000 W)
  PSM 16
    State: Online
    INP0 234 542
    INP1 234 569
    Output 52.49 1043.24
    Capacity 3000 W (maximum 3000 W)
  PSM 17
    State: Online
    INP0 236 586
    INP1 235 556
    Output 52.45 1019.55
    Capacity 3000 W (maximum 3000 W)
  PSM 18
    State: Online
    INP0 236 557
    INP1 235 561
    Output 52.47 1029.89
    Capacity 3000 W (maximum 3000 W)
  PSM 19
    State: Online
    INP0 235 568
    INP1 235 575
    Output 52.51 1056.87
    Capacity 3000 W (maximum 3000 W)

Item Used(W)
  Routing Engine 0 122
  Routing Engine 1 116
  FPC 0 2866
  FPC 1 2226
  FPC 2 3004
  FPC 3 2216
  FPC 4 2173
  FPC 5 1688
  FPC 6 2179
```

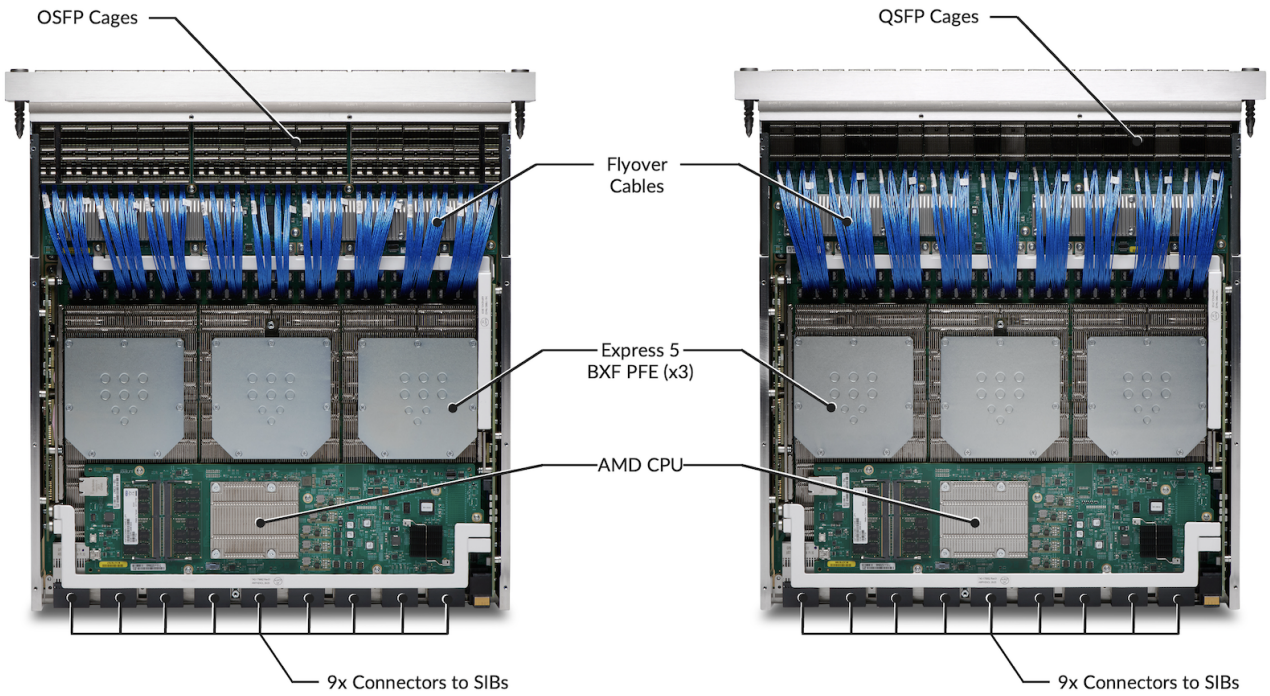
```
    FPC 7 2632
    SIB 0 620
    SIB 1 619
    SIB 2 633
    SIB 3 625
    SIB 4 638
    SIB 5 633
    SIB 6 624
    SIB 7 626
    SIB 8 624
    Fan Tray 0 309
    Fan Tray 1 284
    Fan Tray 2 308

  System:
    Zone 0:
       Capacity: 60000 W (maximum 60000 W)
       Allocated power: 51457 W (8543 W remaining)
       Actual usage: 20760 W
    Total system capacity: 60000 W (maximum 60000 W)
    Total remaining power: 8543 W

  {master}
  root@ptx12008-re0>
```

The same counters are also available through streaming telemetry.

1.16 Line Cards Architecture



The LC54s are based on a simple/optimized architecture in the sense the ports are fixed (no modularity requiring additional mechanical elements, connectors, power distribution, etc.) and their ports are directly connected to the NPU.

Each port is mapped to an individual Port Group, terminating 8x SerDes operated at speeds from 10Gbps (or 25Gbps) to 106Gbps depending on the pluggable present in the optical cage. We don't use any intermediate retimer/ReverseGearBox between the port and the forwarding engine or between the PFE and the fabric connectors.

The only difference between the QSFP and OSFP versions can be found in the mezzanine boards and of course the optical cages. The rest of the card architecture is common to the two types.

They host three BXF instances and each of them contains two datapaths seen as PFE 0 to 5 as shown below.

```
root@ptx12008-re0> show chassis fpc 0 pfe-instance all

FPC 0
PFE-Instance PFE PFE-State
0 0 ONLINE
0 1 ONLINE
1 2 ONLINE
1 3 ONLINE
2 4 ONLINE
2 5 ONLINE

{master}
root@ptx12008-re0>
```

Follow CLI output shows the bandwidth per PFE (7.2Tbps) and also exposes the line card CPU (Snowy Owl AMD 8-core at 2.5GHZ) with 64GB memory.

```
root@ptx12008-re0> show chassis fpc 0 detail
Slot 0 information:
  State Online
  Temperature 51 degrees C / 123 degrees F (BX-0 HBM-0)
  Temperature 52 degrees C / 125 degrees F (BX-0 HBM-1)
  Temperature 50 degrees C / 122 degrees F (BX-1 HBM-0)
  Temperature 49 degrees C / 120 degrees F (BX-1 HBM-1)
  Temperature 52 degrees C / 125 degrees F (BX-2 HBM-0)
  Temperature 52 degrees C / 125 degrees F (BX-2 HBM-1)
  Temperature 49 degrees C / 120 degrees F (CPU)
  Total CPU DRAM 65536 MB
  Start time 2026-02-16 01:45:00 PST
  Uptime 1 day, 9 hours, 48 minutes, 10 seconds
  Max power consumption 4900 Watts
  PFE Type Express-5

PFE Information:

 PFE Power ON/OFF Bandwidth SLC
  0 On 7200
  1 On 7200
  2 On 7200
  3 On 7200
  4 On 7200
  5 On 7200

{master}
root@ptx12008-re0>
```

These line cards are not MPC per say (Modular PIC Concentrators), they are managed by the system as a single PIC, as displayed in the output below.

```
root@ptx12008-re0> show chassis fpc pic-status
Slot 0 Online PTX12K-LC54QDD
  PIC 0 Online PTX12K-54QDD800-LC-PIC
Slot 1 Online PTX12K-LC54OSFP
  PIC 0 Online PTX12K-54OSFP800-LC-PIC

<SNIP>

{master}
root@ptx12008-re0>
```

To check real time utilization of the ingress and egress datapaths

```
root@ptx12008-re0> show chassis utilization forwarding fabric fpc 0
FPC PFE Dir Max Used Free High Watermark Threshold
            Limit Value Timestamp High Low Exceed?
0 0 Ingress 7742 291 7451 291 2026-02-16 17:17:00 95 90 false
0 0 Egress 7742 291 7451 292 2026-02-16 11:30:38 95 90 false
0 1 Ingress 7742 97 7645 97 2026-02-18 01:20:47 95 90 false
0 1 Egress 7742 97 7645 97 2026-02-18 01:20:47 95 90 false
0 2 Ingress 7742 97 7645 97 2026-02-17 19:08:29 95 90 false
0 2 Egress 7742 97 7645 97 2026-02-17 08:50:28 95 90 false
0 3 Ingress 7742 97 7645 97 2026-02-16 23:12:25 95 90 false
0 3 Egress 7742 97 7645 97 2026-02-16 23:12:25 95 90 false
0 4 Ingress 7742 291 7451 291 2026-02-16 04:30:21 95 90 false
```

```
0 4 Egress 7742 291 7451 291 2026-02-16 13:40:01 95 90 false
0 5 Ingress 7742 97 7645 97 2026-02-16 12:22:41 95 90 false
0 5 Egress 7742 97 7645 97 2026-02-16 03:34:41 95 90 false

{master}
root@ptx12008-re0>
```

Note: it's possible to turn off and on one or two PFEs in the LC54 without impacting the service of the remaining one(s). This feature can be leverage in a power saving strategy.

A dedicated article will be published on the power aspect: **the link will be updated here as soon as we will publish it.**

1.17 Interface Configuration and Options

1.18 Port Naming Logic (CIC)

The table below summarizes the interface's naming rules, including channelized ports.

All the ports follow the same rules based on the "Common Interface Configuration" model (CIC), regardless of their position in PICs.

Each physical port is mapped to a unique port group (PG) via 8x SerDes and no intermediate RGB, so we don't have any port combination limitation. You can use all ports with 2x400GigE or 8x100GigE without any constraint. Every port can be configured at the speed you need, but you can only support a unique speed for all members of the same physical port channelized (4x10GigE, 4x25GigE, 8x100GigE, 2x400GigE).

The port numbering on each line card is represented in the Figure 18 and 19 below:

The following illustrations describes the mapping between the ports' position, NPU, and DataPath.

With the bundle PTX12008-BASE (**3x SIB**, 1x RE, 12x PSM), the bandwidth available in each groups of 9 ports displayed above (PFE-0 or PFE-1, blue, green or purple) is reduced to 2.4Tbps total. It's doesn't mean you are limited to 3x 800GbE or 6x 400GbE optics: you can populate all 9 ports of the datapath. But the total bandwidth received or transmitted by all the interfaces of the DP will not exceed 2.4Tbps.

1.19 Ports Capability

The complete list of supported interfaces will be updated soon on the Pathfinder; the following chart provides a couple of examples.

	Port Type	#SerDes and Rate (Gbps)	#SerDes Effective Rate (Gbps)
800GigE	1x 800GAUI-8	8x 106.25	8x 100
400GigE	2x 400GAUI-4	4x 106.25	4x 100
400GigE	1x 400GAUI-8	8x 53.125	8x 50
200GigE	2x 200GAUI-4	4x 53.125	4x 50
100GigE	8x 100GAUI-1	1x 106.25	1x 100
100GigE	2x 100GAUI-4	4x 26.56254x 25.78125	4x 25

	Port Type	#SerDes and Rate (Gbps)	#SerDes Effective Rate (Gbps)
50GigE	2x LAUI-2	2x 25.78125	2x 25
40GigE	1x XLAUI	4x 10.3125	4x 10
25GigE	4x 25GAUI-1	1x 25.78125	1x 25
10GigE	1x XFI	1x 10.3125	1x 10

> Note that we will interchangeably use the “effective bandwidth” (amount of WAN traffic from revenue ports) or the actual bandwidth of the SerDes (106.25/100 or 53.125/50 for example).

As a quick on-box reference, the following CLI command shows transceivers plugged into the port, plus the port speed capabilities (note: it shows capabilities and not necessarily the software support. Please use the port checker and hardware compatibility tools on apps.juniper.net to verify the support).

In a nutshell, QSFP ports can be configured from 10GbE to 800GbE while OSFP can be configured from 100GbE to 800GbE.

QSFP ports capabilities:

```
root@ptx12008-re0> show chassis pic fpc-slot 0 pic-slot 0
FPC slot 0, PIC slot 0 information:
  Type PTX12K-54QDD800-LC-PIC
  State Online
  PIC version 255.255
  Uptime 1 day, 9 hours, 44 minutes, 27 seconds

PIC port information:
                    Fiber Xcvr vendor Wave- Xcvr JNPR MSA Short/Long
  Port Cable type type Xcvr vendor part number length Firmware Rev Version Port
  0 800G-AOC 10M MM JUNIPER-1W1 740-150874 1311 nm 3.6.00 REV 01 CMIS 5.1 L

<SNIP>

  52 800G-AOC 10M MM JUNIPER-1W1 740-150874 1311 nm 3.6.00 REV 01 CMIS 5.1 L
  53 800G-AOC 10M MM JUNIPER-1W1 740-150874 1311 nm 3.6.00 REV 01 CMIS 5.1 L

Port speed information:

  Port PFE Capable Port Speeds
  0 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  1 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  2 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  3 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  4 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  5 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  6 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  7 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  8 0 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  9 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G
       5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  10 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
      x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  11 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
      x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
  12 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
      x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
```

```
13 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
14 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
15 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
16 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
17 1 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
18 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
19 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
20 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
21 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
22 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
23 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
24 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
25 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
26 2 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
27 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
28 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
29 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
30 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
31 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
32 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
33 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
34 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
35 3 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
36 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
37 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
38 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
39 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
40 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
41 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
42 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
43 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
44 4 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
45 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
46 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
47 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
48 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
49 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
50 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
51 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
52 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
    x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G
53 5 1x10G 4x10G 4x25G 8x25G 1x40G 4x50G 8x50G 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1
```

```
        x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G 1x400G 2x400G 1x800G

{master}
root@ptx12008-re0>
```

OSFP ports capabilities:

```
root@ptx12008-re0> show chassis pic fpc-slot 7 pic-slot 0
FPC slot 7, PIC slot 0 information:
  Type PTX12K-54OSFP800-LC-PIC
  State Online
  PIC version 255.255
  Uptime 1 day, 9 hours, 43 minutes, 53 seconds

PIC port information:
                    Fiber Xcvr vendor Wave- Xcvr JNPR MSA Short/Long
  Port Cable type type Xcvr vendor part number length Firmware Rev

  Version Port
  0 OSFP-PASSIVE-E-LPBK n/a MULTILANE ML4064-LB112-30W n/a 1.0 XXXX CMIS 2.8 NA

<SNIP>

Port speed information:

  Port PFE Capable Port Speeds
  0 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  1 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  2 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  3 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  4 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  5 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  6 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  7 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  8 0 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  9 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4x200G
       2x400G 1x800G
  10 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  11 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  12 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  13 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  14 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  15 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  16 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  17 1 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  18 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  19 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  20 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  21 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  22 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  23 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  24 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  25 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
  26 2 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
       x200G 2x400G 1x800G
```

```
47    27 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
48    28 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
49    29 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
50    30 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
51    31 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
52    32 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
53    33 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
54    34 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
55    35 3 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
56    36 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
57    37 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
58    38 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
59    39 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
60    40 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
61    41 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
62    42 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
63    43 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
64    44 4 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
65    45 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
66    46 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
67    47 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
68    48 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
69    49 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
70    50 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
71    51 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
72    52 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
73    53 5 1x100G 2x100G 3x100G 4x100G 1x200G 2x200G 1x400G 5x100G 6x100G 7x100G 8x100G 3x200G 4
          x200G 2x400G 1x800G
74
75  {master}
76  root@ptx12008-re0>
```

1.20 Conclusion

This article, despite its length, only scratched the surface of the PTX12000 chassis. We explored at high level the various FRUs, the interface configuration and some basic management functions. Many more articles will come complete it in the next weeks. Stay tuned…

1.21 Useful links

- Introduction to Express 5 TechPost blog: https://techpost.hpe.com/articles/express-5-overview/article
- PTX12000 Chassis Datasheet: https://www.hpe.com/psnow/doc/a00155677enw?jumpid=in_pdfviewer-psnow

- Rethinking Fabric Redundancy: https://techpost.hpe.com/articles/rethinking-fabric-redundancy/article
- Deepdive in the Fabric Topology: https://techpost.hpe.com/articles/ptx12008-fabric-topology/article
- Fabric Self Healing and Recovery: https://techpost.hpe.com/articles/self-healing-and-recovery-mechanisms-for-ptx12008/article
- Navigating the Front Panel Display: https://techpost.hpe.com/articles/navigating-the-ptx12008-lcd-module/article
- PTX12k Power Saving and Metrics (coming soon)

1.22 Glossary

- ASIC: Application Specific Integrated Circuit
- BIER: Multicast using Bit Index Explicit Replication
- BF: Express 5 Package with only F-chiplet and used in the SF5 cards
- BXF: Express 5 Package made of one X-chiplet for WAN connectivity and one F-chiplet for fabric connectivity
- CIC: Common Interface Configuration
- CLI: Command Line Interface
- CPU: Central Processor Unit
- DCI: DataCenter Interconnect
- DRAM: Dynamic Random Access Memory
- EMI: Electromagnetic Interference
- FIB: Forwarding Information Base
- FPGA: Field Programmable Gate Arrays
- FPC: Flexible PIC Concentrator
- FRU: Field Replacement Unit
- GigE: Gigabit Ethernet
- HBM: High-Bandwidth Memory
- MACsec: Media Access Control Security
- OD: Orthogonal Direct
- OSFP: Octal Small Form-factor Pluggable
- PIC: Port Interface Card
- PSM/PSU: Power Supply Module/Unit
- PTP: Precision Time Protocol
- QDD: QSFP Double Density
- RCB: Routing and Control Board
- RE: Routing Engine
- RGB: Reverse GearBox
- RIB: Routing Information Base
- SerDes: Serializer/Deserializer
- SIB: Switch Interface Board
- Sync-E: Synchronous Ethernet
- XSR: Extra Short Reach (SerDes)

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- Kashif Nawaz
- Santo K
- Rahul Kulkarni
- Gurmeet Singh
- Girish Dadhich
- Rahul Unnikrishnan



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